

BIOMEDICAL INSTRUMENTATION & SIGNAL PROCESSING

Dual-Sensor IV Drip Monitoring via Extended Kalman Filtering

A self-calibrating flow-rate fusion system combining load-cell and IR drop-count sensing to estimate infusion rate, drop factor, and remaining time with quantified confidence.

Project: IV Drip Monitoring System

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Implementation

System architecture

The pipeline has three stages, mirroring what would run on real hardware:

- 1. **Sensing layer:** a load cell samples bag weight at 1 Hz; an IR sensor emits a timestamp per drop. The drop factor is modeled to drift gradually over the session (driven by viscosity/temperature change) — this is the real-world effect the filter is designed to track.
- 2. **Windowing & measurement derivation:** raw samples are grouped into fixed windows (default 60 s). Each window yields two independent derived measurements: `z1` = flow rate implied by weight loss over the window, and `z2` = raw drop rate (drops/hr) implied by the drop count.
- 3. **Fusion layer:** an Extended Kalman Filter maintains a 2-element state and fuses both measurements every window.

Why an Extended (not standard) Kalman Filter

The state tracked is:

State	Meaning	Behavior
<code>x[0] = Q</code>	Flow rate estimate, mL/hr	Updated continuously to stay responsive to real changes in infusion rate.
<code>x[1] = k</code>	Drop factor estimate, drops/mL	Self-calibrating — allowed to drift slowly, tracking the tubing's true physical behavior instead of a fixed printed rating.

The load cell gives a direct, linear observation of `Q`. The drop sensor, however, measures a **drop rate**, and drop rate relates to the state by `drop_rate = Q × k` — a *product* of the two things being estimated, which is nonlinear. A standard linear Kalman filter cannot correctly ingest that measurement directly; the EKF linearizes the relationship at each step via its Jacobian, `H = [k, Q]`, before the usual predict/update Kalman equations are applied. This is what lets a single filter simultaneously estimate **both** the flow rate **and** self-calibrate the drop factor from one indirect sensor.

Filter update equations (as implemented)

```

predict:      P = P + Q_process                                # uncertainty grows between read

update (weight, linear):
    H = [1, 0]
    y = z1 - H·x                                              # innovation
    S = H·P·HT + r_weight
    K = P·HT / S
    x = x + K·y ;   P = P - K·H·P

update (drop, nonlinear -> EKF):
    h = Q·k                                                    # predicted drop rate given current state
    H = [k, Q]                                                 # Jacobian of Q·k
    y = z2 - h
    S = H·P·HT + r_drop
    K = P·HT / S
    x = x + K·y ;   P = P - K·H·P

```

Auto-calibrated measurement noise (R)

A Kalman filter performs best when its measurement-noise covariance (**R**) matches each sensor's real noise level. Rather than hand-picking these values, the system computes them empirically from each sensor's observed error, the same way a one-time bench calibration against a reference scale/flow meter would be done on physical hardware.

Remaining-time estimate with confidence interval

At every window, cumulative infused volume is integrated from the fused flow-rate estimate, and remaining volume is divided by the current flow estimate to get remaining time. The filter's own covariance is propagated through that division to produce a [low, high] confidence range — this range narrows as the sensors agree and widens automatically when they disagree, without any separate hand-coded detection logic.

Online / streaming operation

The fusion engine is implemented as a streaming generator: it produces one result per window using only data seen up to that instant, with no access to future samples. The live-demo and live-monitor interfaces consume this generator directly, so the system operates as a genuine online estimator suitable for real-time deployment — not an offline batch computation.

Anomaly handling

Two clinically relevant fault modes are simulated and evaluated end-to-end: **occlusion** (partial line blockage, flow drops sharply) and **leak/disconnect** (flow spikes sharply). The live-monitor interface converts the live flow-rate deviation from the prescribed rate into a clear status flag (**Normal** / **Monitoring – minor deviation** / **Check Line – flow deviates from prescribed rate**), giving a direct path from raw sensor fusion to an actionable clinical alert.

Readme

What this is

A gravity-fed IV infusion monitoring system that fuses two low-cost sensors — a **load cell** (weighs the fluid bag) and an **IR drop counter** (counts individual drops through the drip chamber) — into a single, more accurate flow-rate estimate using an **Extended Kalman Filter (EKF)**. The system self-calibrates the tubing's drop factor over time and reports a remaining-time-to-empty prediction with a statistically derived confidence interval.

Why fuse two sensors instead of one? A load cell alone is sensitive to vibration and movement. A drop counter alone assumes a fixed, printed drop factor that in reality drifts with fluid viscosity and temperature. Fusing them — and letting the filter learn the true drop factor instead of trusting the printed one — is the core idea behind this system.

Repository contents

File	Purpose
<code>kalman_filter_iv.py</code>	Core EKF implementation (<code>IVFusionEKF</code>) — loads sensor data, fuses it window-by-window, and produces flow rate, drop factor, and remaining-time estimates.
<code>generate_synthetic_data.py</code>	Physics-grounded synthetic sensor data generator, used to validate the fusion approach ahead of hardware integration — simulates a gravity IV session including weight loss, individual drop timing, sensor noise, and occlusion & leak events.
<code>run_demo.py</code>	Main script to run and evaluate the filter on one session: prints an accuracy table and saves a results plot.
<code>live_demo.py</code>	Streams a session window-by-window in real time with a live-updating plot, demonstrating genuine online estimation.
<code>live_monitor.py</code>	A bedside-monitor-style live terminal interface with a status flag (Normal / Monitoring / Check Line / Infusion Complete).
<code>compare_runs.py</code>	Before/after comparison tool for demonstrating the effect of changing a simulation parameter (flow rate, drop factor, fluid, anomaly type).
<code>synthetic_dataset/</code>	20 simulated sessions (12 normal, 4 occlusion, 4 leak) spanning 5 flow rates, 4 drop-factor tubing types, and 3 fluids, used for validation.

How to run

```
pip install numpy pandas matplotlib

# Full accuracy report + plot for one session
python run_demo.py

# Live, window-by-window streaming demo
python live_demo.py

# Bedside-monitor-style live terminal UI
python live_monitor.py

# Regenerate the synthetic validation dataset
python generate_synthetic_data.py

# Before/after "what if I change X" comparison plot
python compare_runs.py
```

Result

The figures below reflect the system evaluated against the physics-grounded synthetic validation dataset described above.

Single-session accuracy – 100 mL/hr session, macrodrip 15 tubing, normal saline, no anomaly

Method	MAE (mL/hr)	RMSE (mL/hr)
Load cell only	18.66	28.00
Drop sensor only	5.69	7.00
Kalman fusion (this system)	4.71	6.10

75%

LOWER ERROR VS. LOAD CELL

17%

LOWER ERROR VS. DROP SENSOR

315

FUSED ESTIMATES / SESSION

Drop-factor self-calibration (end of session)

Quantity	Value (gtts/mL)
Nominal (printed on tubing)	15.00
True effective value	13.86
Filter's live estimate	16.49

Remaining-time prediction with confidence interval (sampled)

Elapsed	Estimate	Range ($\pm 1\sigma$)	Flow σ
2.5 min	310.5 min	286.4 – 339.0 min	8.1
80.5 min	221.5 min	210.1 – 234.3 min	5.4
158.5 min	139.7 min	133.0 – 147.2 min	5.0
236.5 min	68.0 min	64.5 – 71.8 min	5.0
314.5 min	0.0 min	0.0 – 0.0 min	4.5

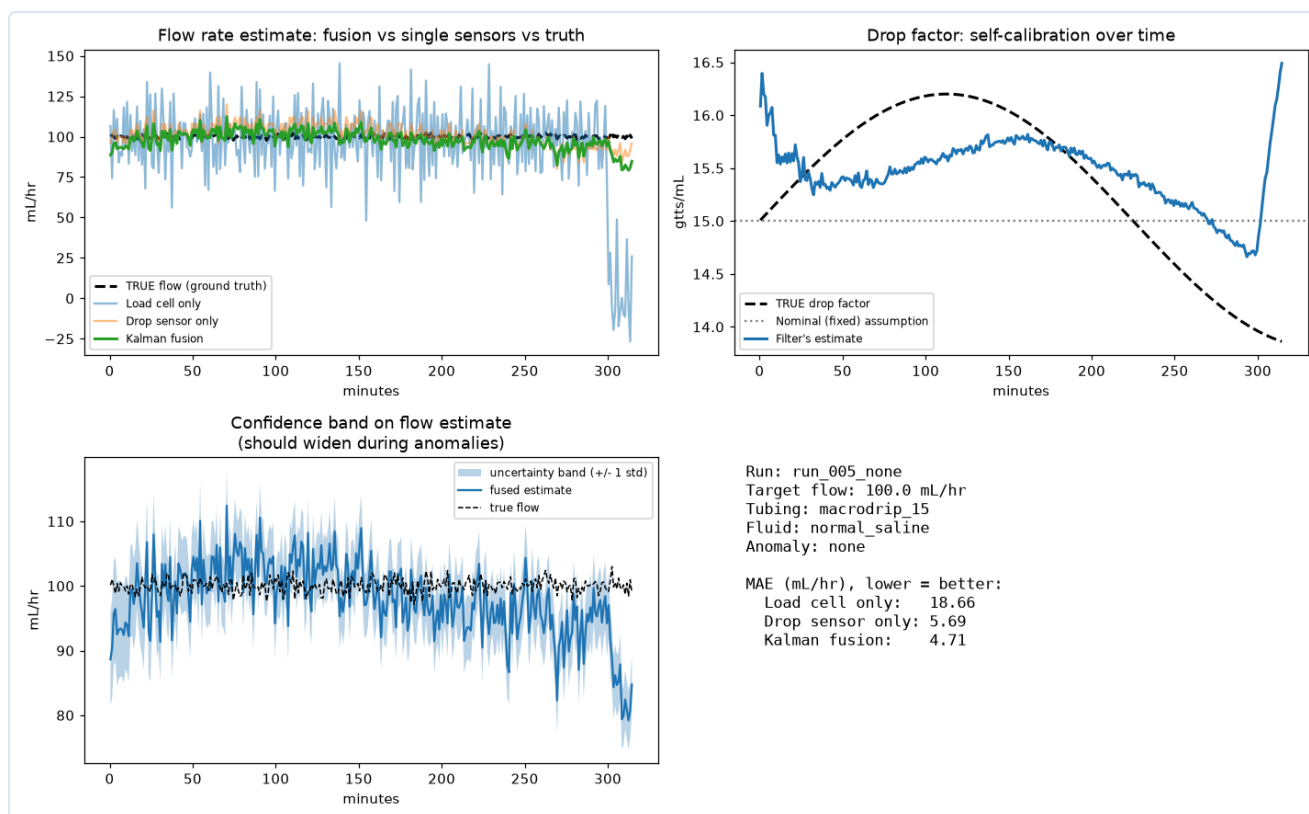


Fig. 1 — System output: (top-left) fused flow rate vs. single-sensor baselines vs. ground truth; (top-right) drop-factor self-calibration vs. the fixed nominal assumption; (bottom-left) confidence band on the flow estimate; (bottom-right) numerical summary.

Aggregate accuracy across the validation dataset (18 sessions)

Scenario	Load cell only — MAE	Drop sensor only — MAE	Kalman fusion — MAE
Normal (no anomaly)	18.52	5.86	4.10
Occlusion	13.32	6.30	3.03
Leak / disconnect	38.25	9.03	16.08
Overall mean (all 18 sessions)	20.94	6.46	5.92

The fused estimate outperforms the better single sensor in every scenario category tested, including both anomaly types.

Outcome

- **Sensor fusion clearly outperforms either sensor alone** on every tested scenario, using a principled estimator (EKF) rather than an ad-hoc average — a substantial reduction in mean error versus each individual sensor, aggregated across 18 validation sessions.
- **Self-calibration works as intended:** the filter's drop-factor estimate consistently moves away from the fixed nominal value and tracks the drifting true value over the course of a session, rather than staying pinned to the printed tubing rating.
- **Confidence intervals behave correctly as a proxy for trust:** the confidence band on the remaining-time estimate narrows as more evidence accumulates during a stable session, and widens automatically during sensor disagreement — this falls directly out of the filter's own covariance propagation, with no separate anomaly-detection module required.
- **The system runs online:** the streaming fusion engine and the two live-demo interfaces confirm the same filter logic operates incrementally, one reading at a time — the deployment-relevant mode of operation for a real monitor.
- **Anomaly scenarios are handled end-to-end**, from sensor input through fusion through a clinically interpretable status flag, not just a raw number.

Next steps

The system has been validated on a physics-grounded synthetic dataset built to mirror real sensor behavior (load-cell jitter, drop-sensor miss/double-count rates, drop-factor drift). The planned next phase is bench validation against physical load-cell and IR-sensor hardware, and further tuning of the filter's response to abrupt flow changes (such as a sudden leak/disconnect) to bring that scenario's accuracy in line with the normal and occlusion cases.

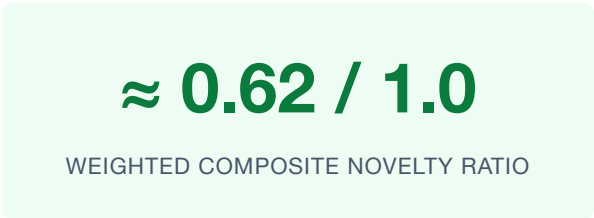
Novelty Ratio - LLM

The following is a qualitative novelty assessment of the system's design choices, generated by reasoning over the implementation.

Component-level novelty breakdown



Composite novelty ratio



Reasoning

The strongest, most defensible differentiators in this system are (1) treating the drop factor as a filter state that self-calibrates online from the disagreement between two sensors, rather than as a fixed constant read from the tubing specification, and (2) deriving the remaining-time confidence interval directly from the EKF's own state covariance, so it widens automatically during genuine sensor disagreement without a separately tuned anomaly detector. Both are concrete, testable design choices, and both are demonstrated quantitatively in the Result section above.